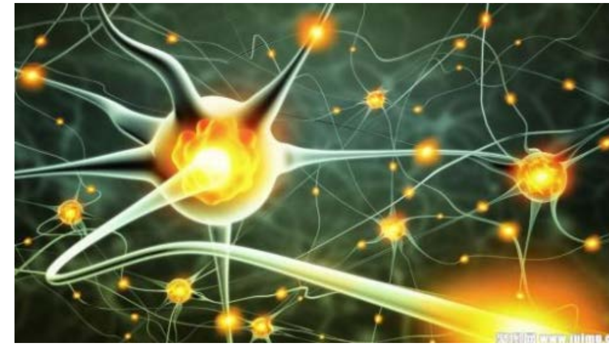
  

Chapter 2
Basic Neuroscience

Minghui Shi

Xiamen University, China

Processing Stages --- Review

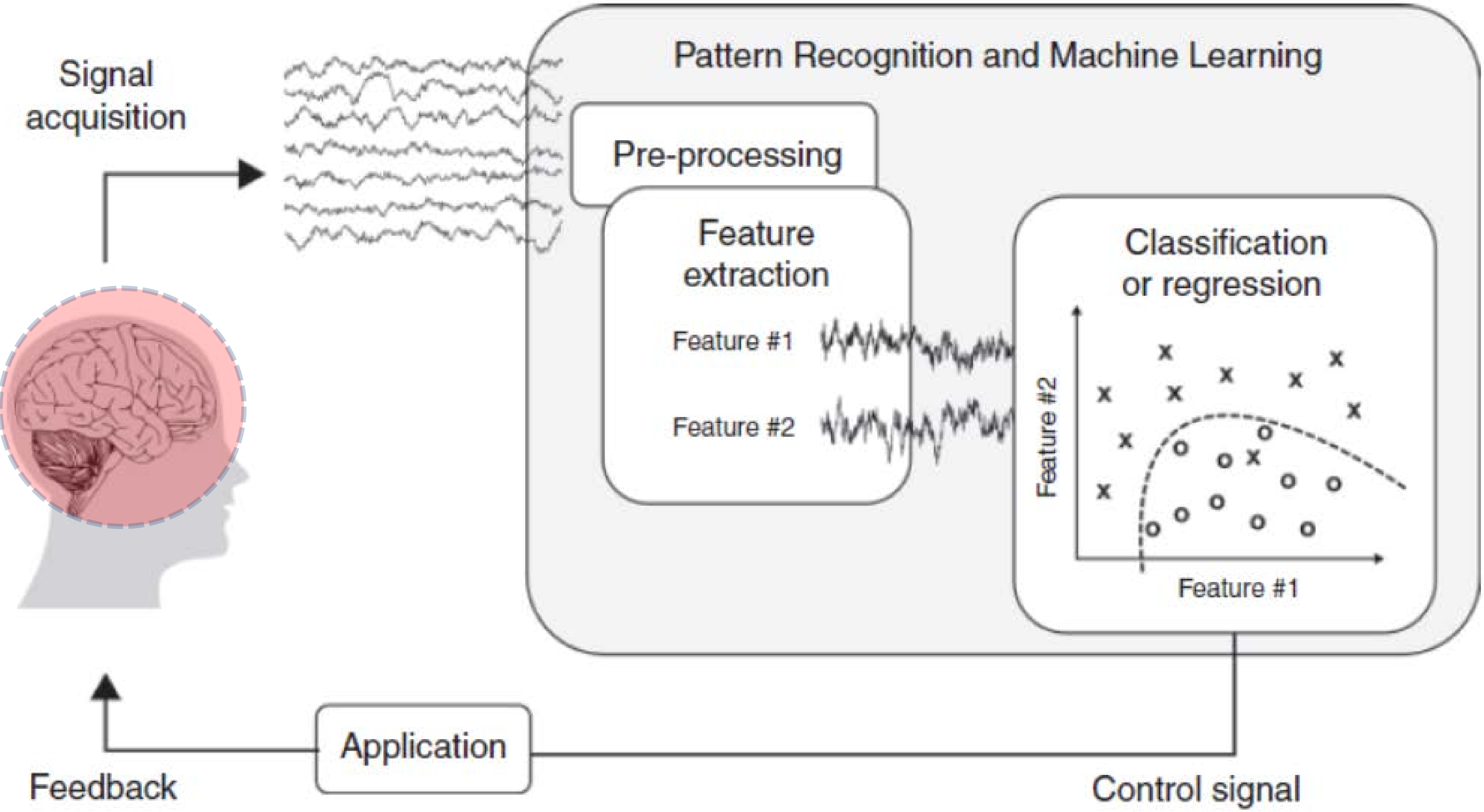


Figure 1.1. **Basic components of a brain-computer interface (BCI).**

Chapter 2 Basic Neuroscience

Content

2.0 Brain Function

- What is the average weight of the human brain?
- How does the brain enact a behavior?
- What are the characteristics of the computing model of the brain?
- What's the computing unit of brain?
- What is the difference of the computing model between the brain and traditional computers?
- Can Computer V.S. Brain? Why?

Relevant Video

- Video 1 (5:50-8:30)
- Video 2 (7:00-8:00)
- Video 3 (0:00-1:00)
- Video 4 (1:50-5:50 min)

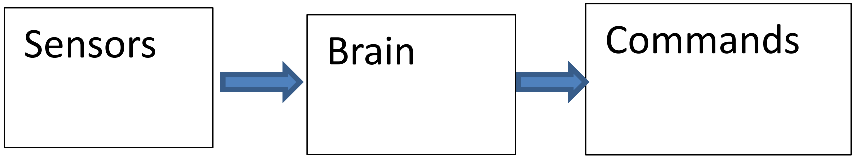
2.0 Brain Function

What's the weight of a brain?

A: About three pounds.

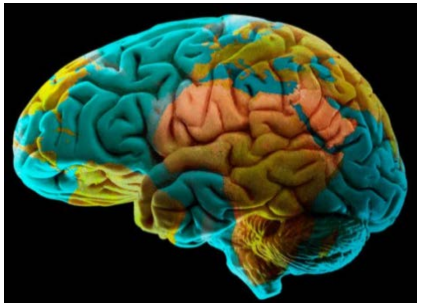
How does the brain enact a behavior?

A: The brain transforms signals from millions of sensors located all over the body into appropriate muscle commands to enact a behavior suitable to the task at hand.

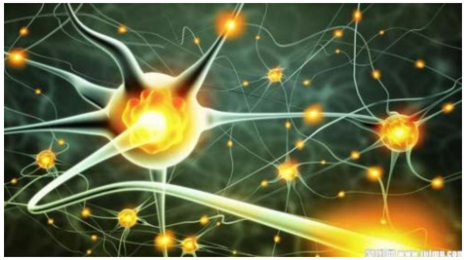


2.0 Brain Function

- **What are the characteristics of the computing model of the brain?**
  - Parallel
  - Distribute
  - Plastic



2.0 Brain Function



- **What's the computing unit of brain?**
- A: The workhorse of the brain is a type of cell known as a **neuron**, a complex electrochemical device that
  - **receives** information from hundreds of other neurons.
  - **processes** this information.
  - **conveys** its output to hundreds of other neurons.

2.0 Brain Function

- **What's the main difference of computing model between brain and computer?**
  - The connections between neurons are **plastic**, allowing the brain's networks to adapt to new inputs and changing circumstances.
  - This adaptive and distributed mode of computation **sets** the brain **apart from** traditional computers, which are based on the **von Neumann architecture** with a separate central processing unit, memory units, fixed connections between components, and a serial mode of computation.

2.0 Brain Function

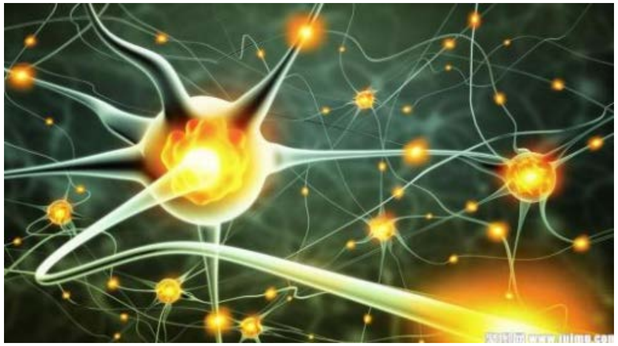
- **Can Computer V.S. Brain ?**
  – This closed-loop, real-time control system remains unsurpassed by any artificially created system despite decades of attempts by computer scientists and engineers.
- **Why?**
  – The brain's unique information processing capabilities arise from its massively parallel and distributed way of computing.

Outline

- In this chapter, we explore:
  - **Biophysical properties of neurons.**
  - How is the electricity in the brain generated?
  - **How neurons communicate with each other?**
  - **How neurons transmit information to other neurons?**
  - **How synapse are adapted in response to inputs and outputs?**
  - **Neural network architecture**
  - **Anatomy of the brain**

2.1 Neurons

- A neuron is a type of cell that is generally regarded as the basic computational unit of the nervous system.



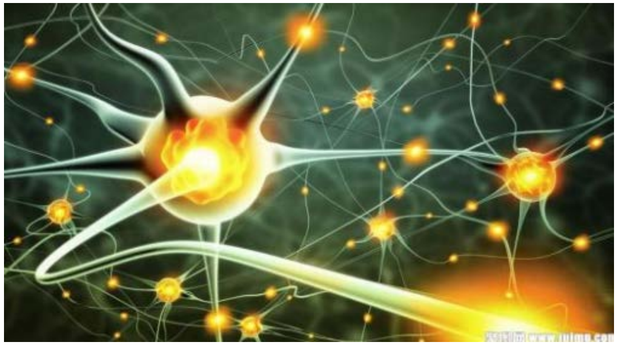
2.1 Neurons

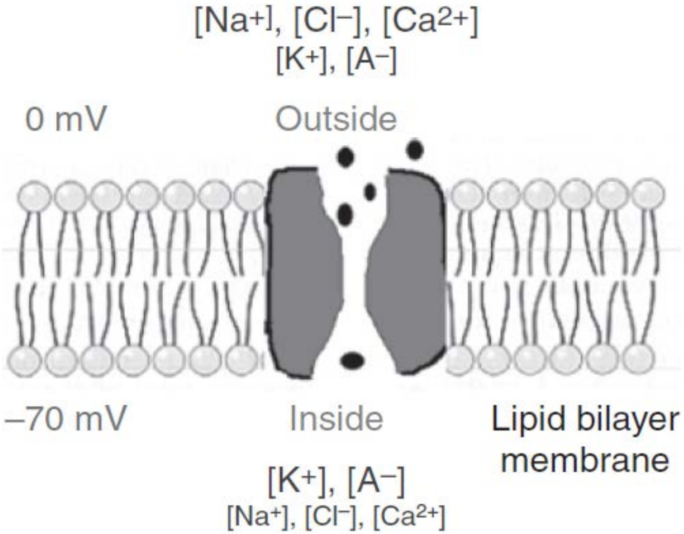
How is the electricity in the brain generated?

（Rest Potential）

- As a crude approximation, the neuron can be regarded as a leaky bag of charged liquid.

- The membrane of a neuron is made up of a lipid bi-layer that is impermeable except for **openings** called **ionic channels** that selectively allow the passage of particular kinds of ions.





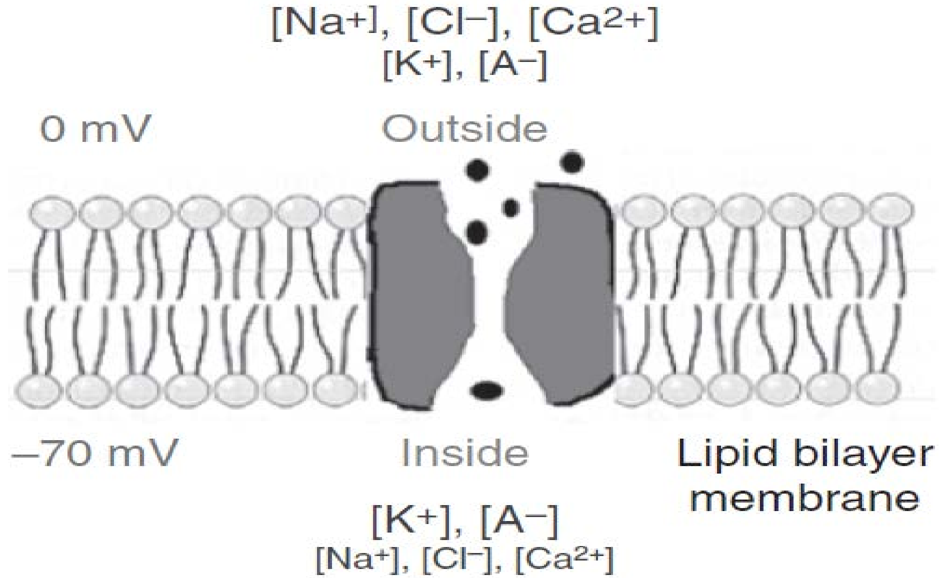


Figure 2.1. **The electrochemical dance of ions in a neuron.**

Neurons reside in an aqueous medium with a larger concentration of sodium (Na^+), chloride (Cl^-), and calcium (Ca^{2+}) on the outside of the cell， and a greater concentration of potassium (K^+) and organic anions (A^-) inside the cells.

As a result of this imbalance, **there is a potential difference of approximately – 65 to – 70 mV** across the neuron’s membrane when the neuron is at rest. There exist active pumps that work to maintain this potential difference by expending energy.

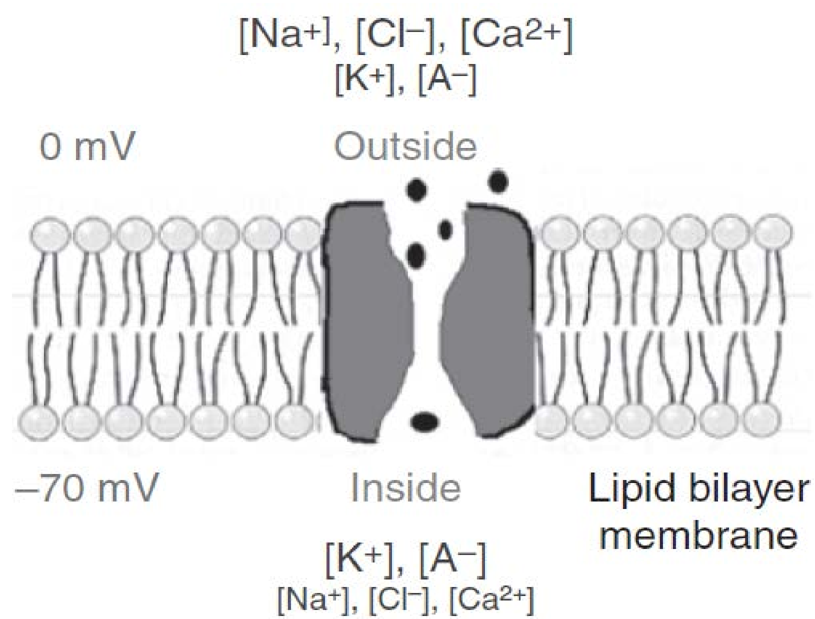
2.2 Action Potentials or Spikes

How is the electricity in the brain generated?

（Action Potential）

- When the neuron receives **sufficiently strong inputs** from other neurons, a cascade of events is triggered, resulting in **rapid rise and fall** of the membrane potential.
(see detail at page 8)

- The rapid rise and fall of the membrane potential is called an **action potential** or **spike**, and represents the **dominant mode of communication** between one neuron and another.

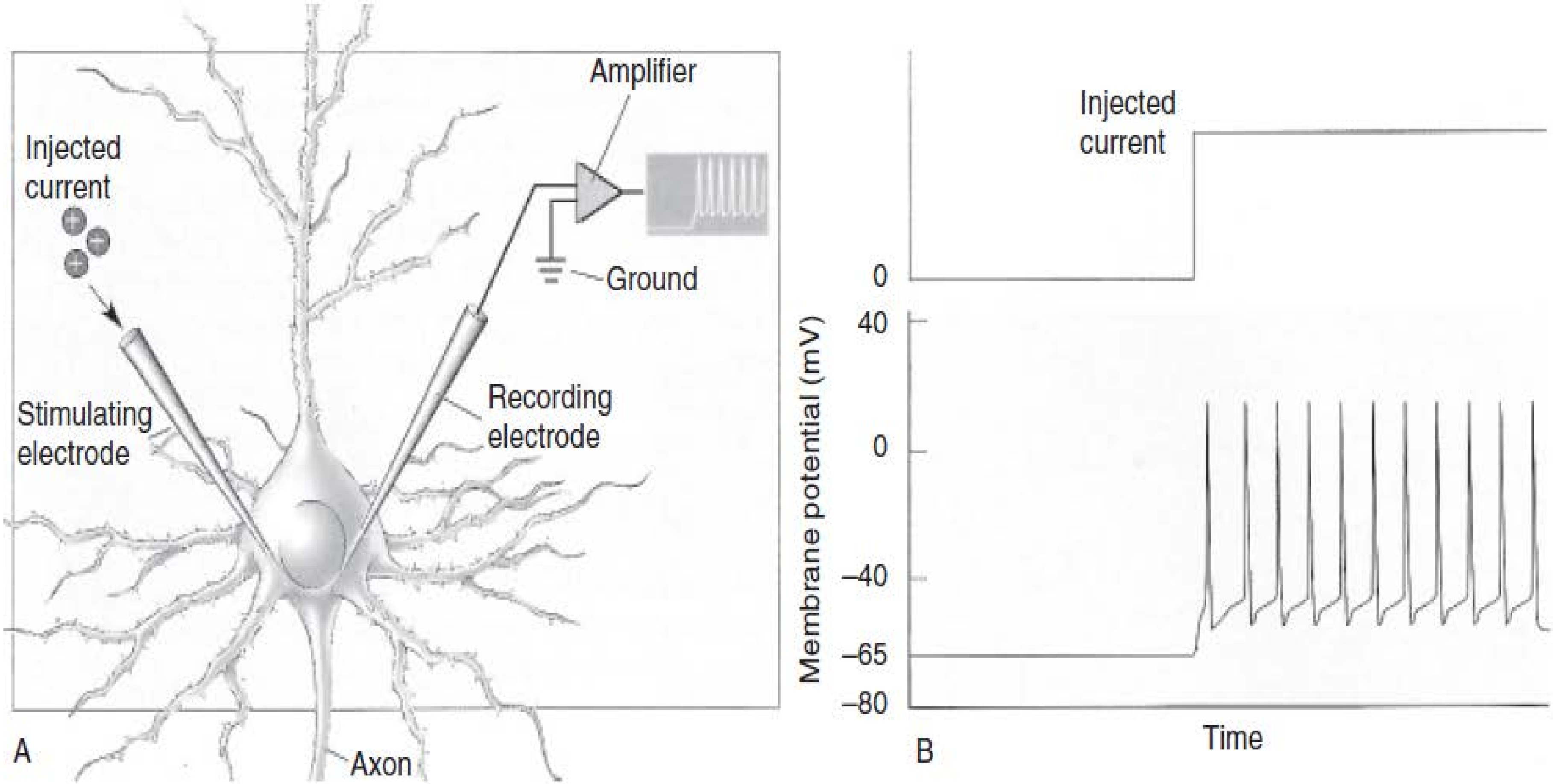


2.2 Action Potentials or Spikes

- The spike is an all-or-none stereotyped event with **little** or **no information in the shape of the spike itself** – **information** is thought to be conveyed instead by the **firing rate** (number of spikes per second) and/or **the timing of spikes**.

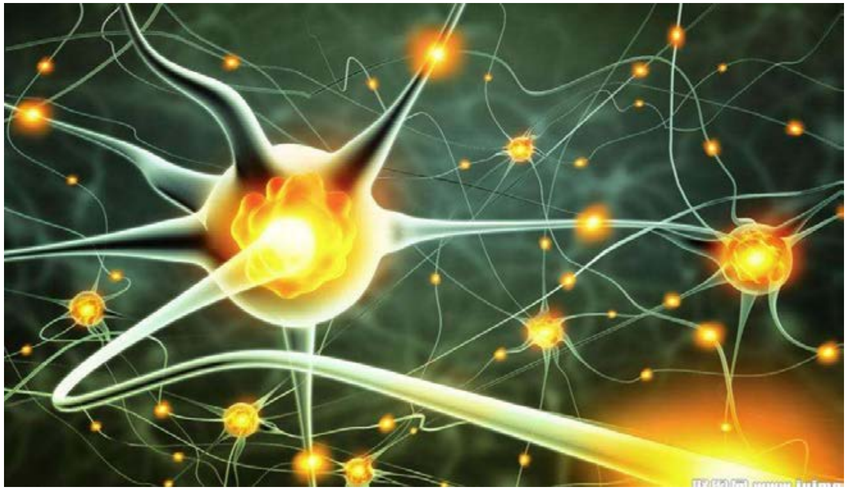
- Neurons are therefore often modeled as emitting a 0 or 1 digital output.

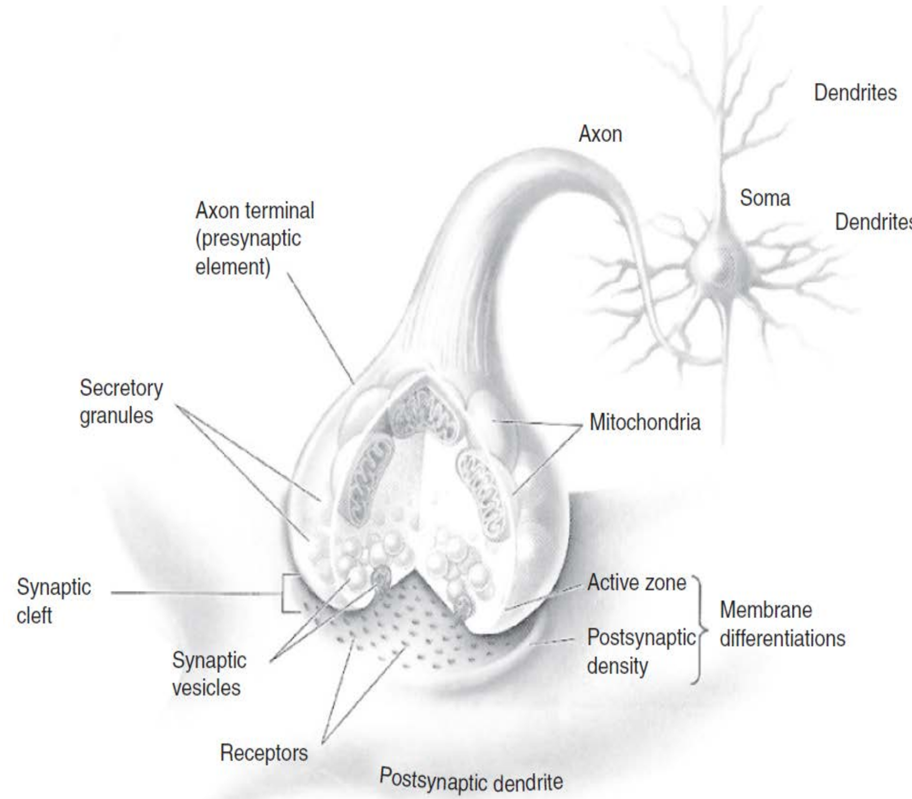
2.2 Action Potentials or Spikes

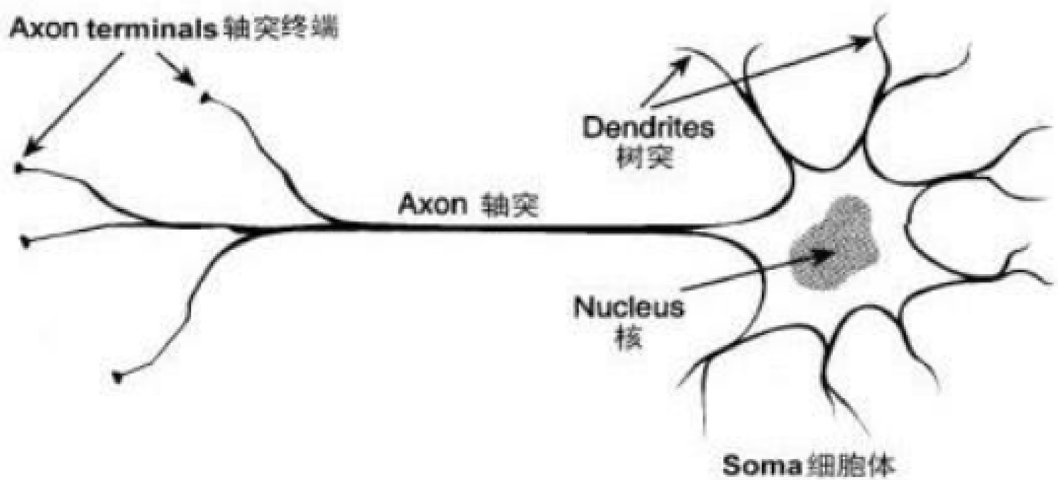


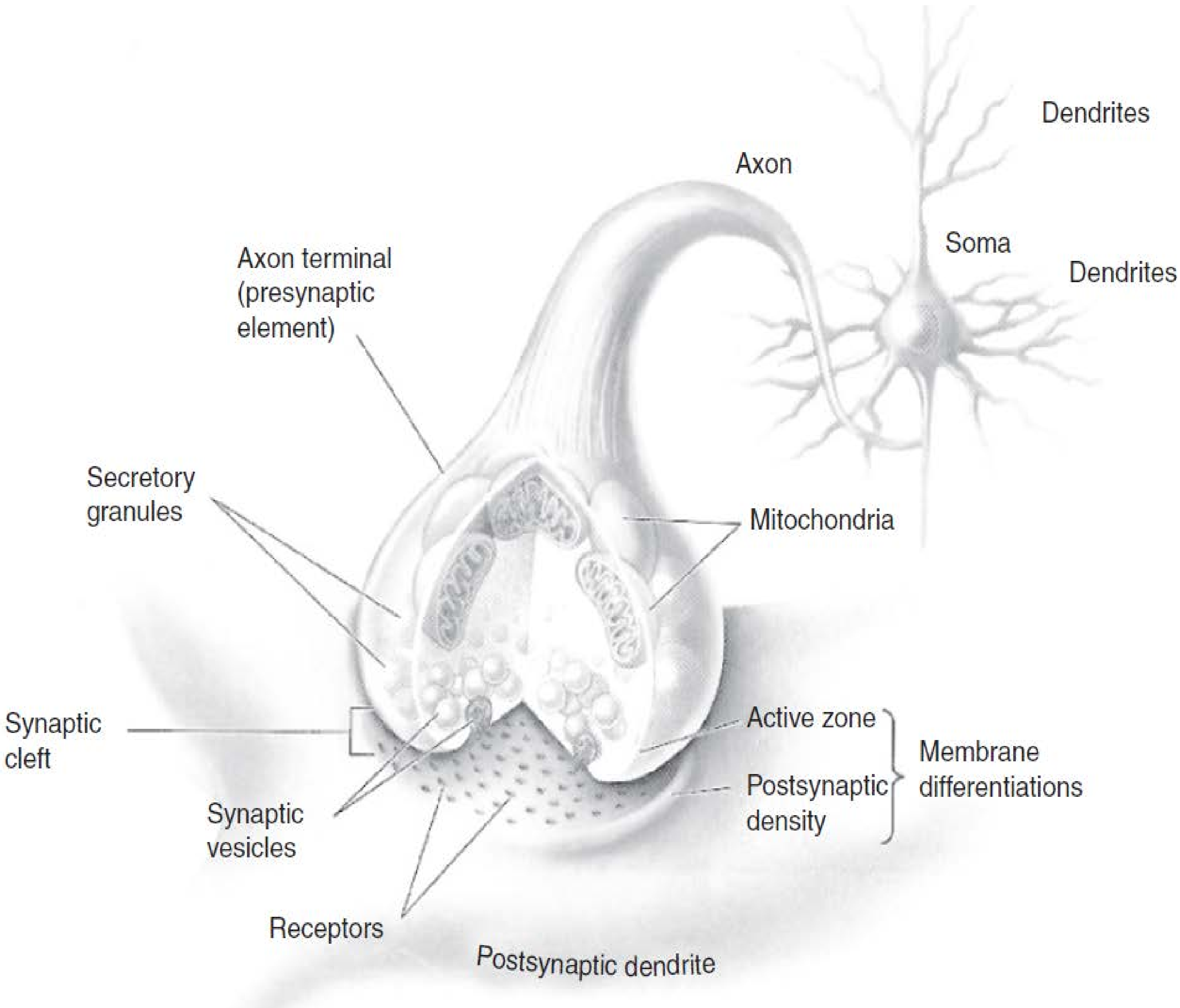
http://open.163.com/movie/2009/11/P/B/M6I12OHDE_M6I3L43PB.html
（6:36-8:00）

2.3 Dendrites and Axons



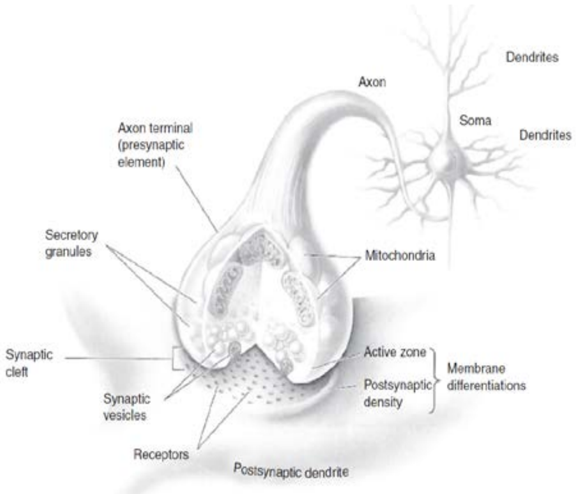


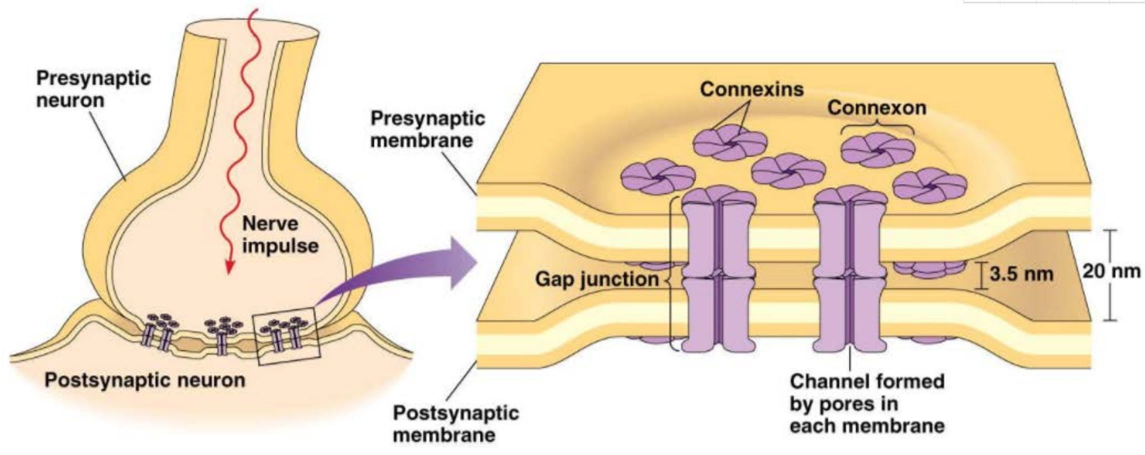




Dendrites
Axon
Soma
Dendrites
Axon terminal
(presynaptic
element)
Secretory
granules
Mitochondria
Synaptic
cleft
Active zone
Membrane
differentiations
Postsynaptic
density
Synaptic
vesicles
Receptors
Postsynaptic dendrite

2.4 Synapses (How neurons communicate with each other?)





(a) An electrical synapse (b) Gap junctions

Neurons communicate with each other through connections known as synapses. .

Synapses can be electrical but are more typically chemical.

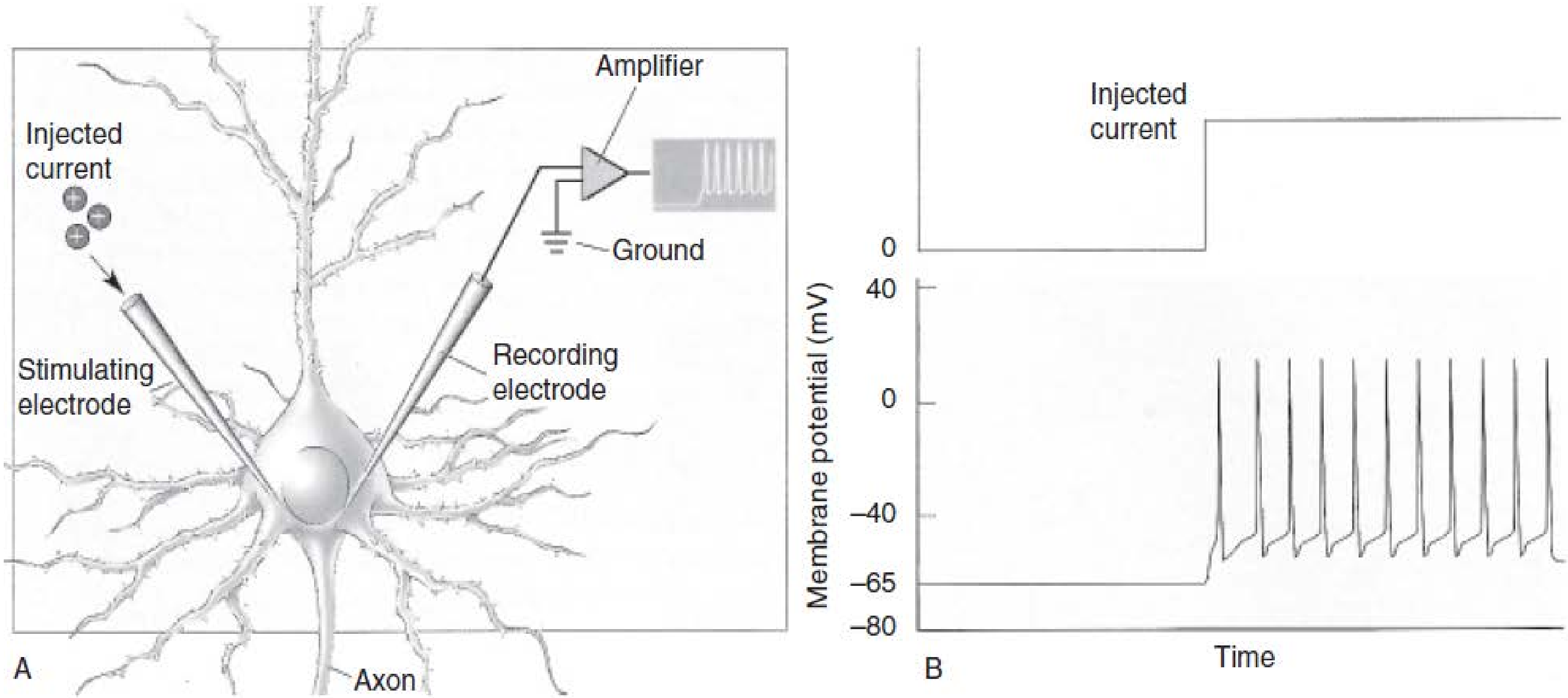
A synapse is essentially a gap or cleft between the axon of one neuron (called the presynaptic neuron) and a dendrite
(or soma) of another neuron (called the postsynaptic neuron).

2.5 Spike Generation

- Video 1 (5:50-8:30)
- Video 2 (7:00-8:00)
- Video 3 (0:00-1:00)
- Video 4 (1:50-5:50 min)

2.5 Spike Generation

- **A simple threshold model of spike generation**



2.6 Adapting the Connections: Synaptic Plasticity

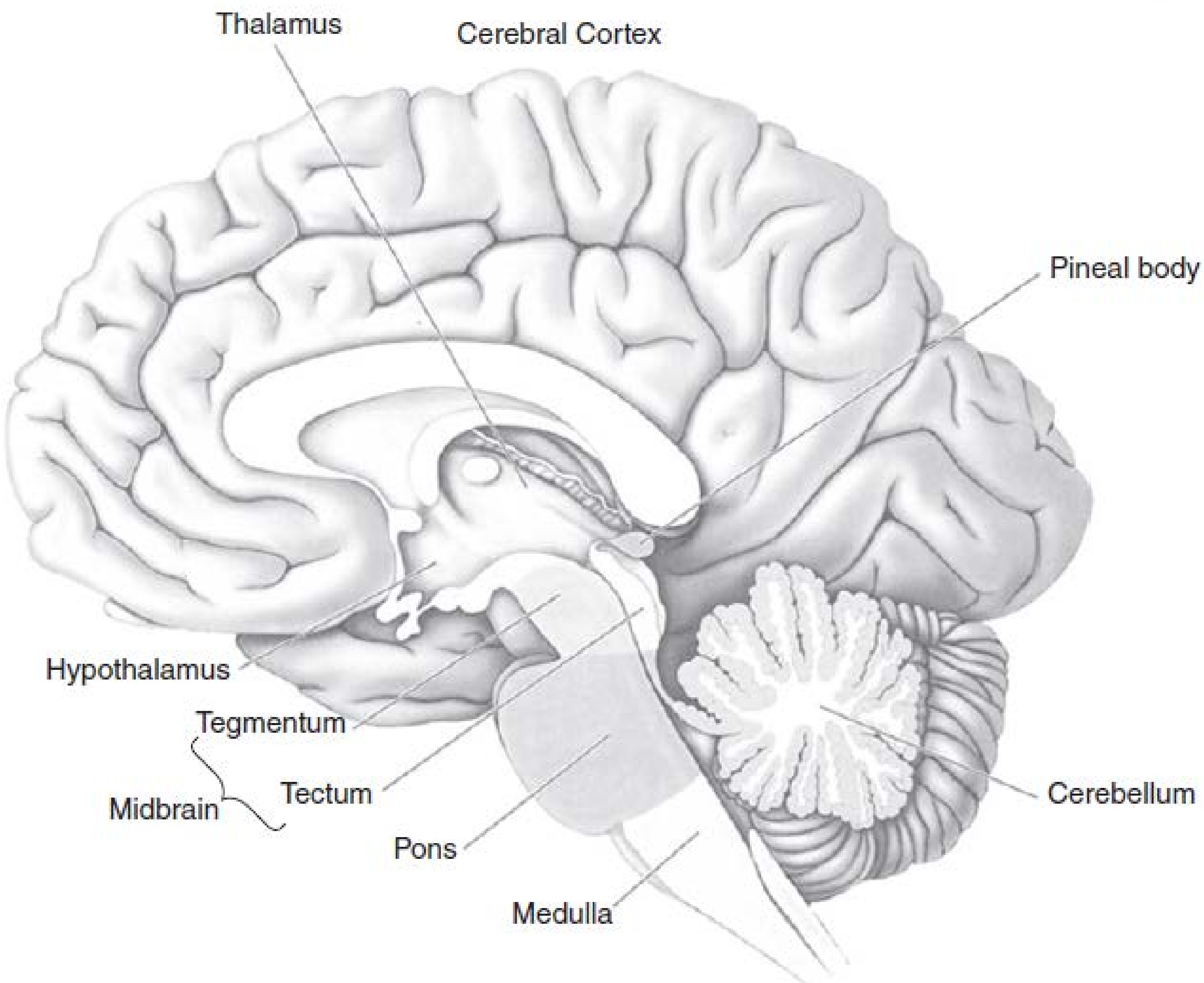
- **Synaptic plasticity**
- A critical component of the brain's adaptive capabilities is the ability of neurons to change the strength of the connections between themselves through **synaptic plasticity**.

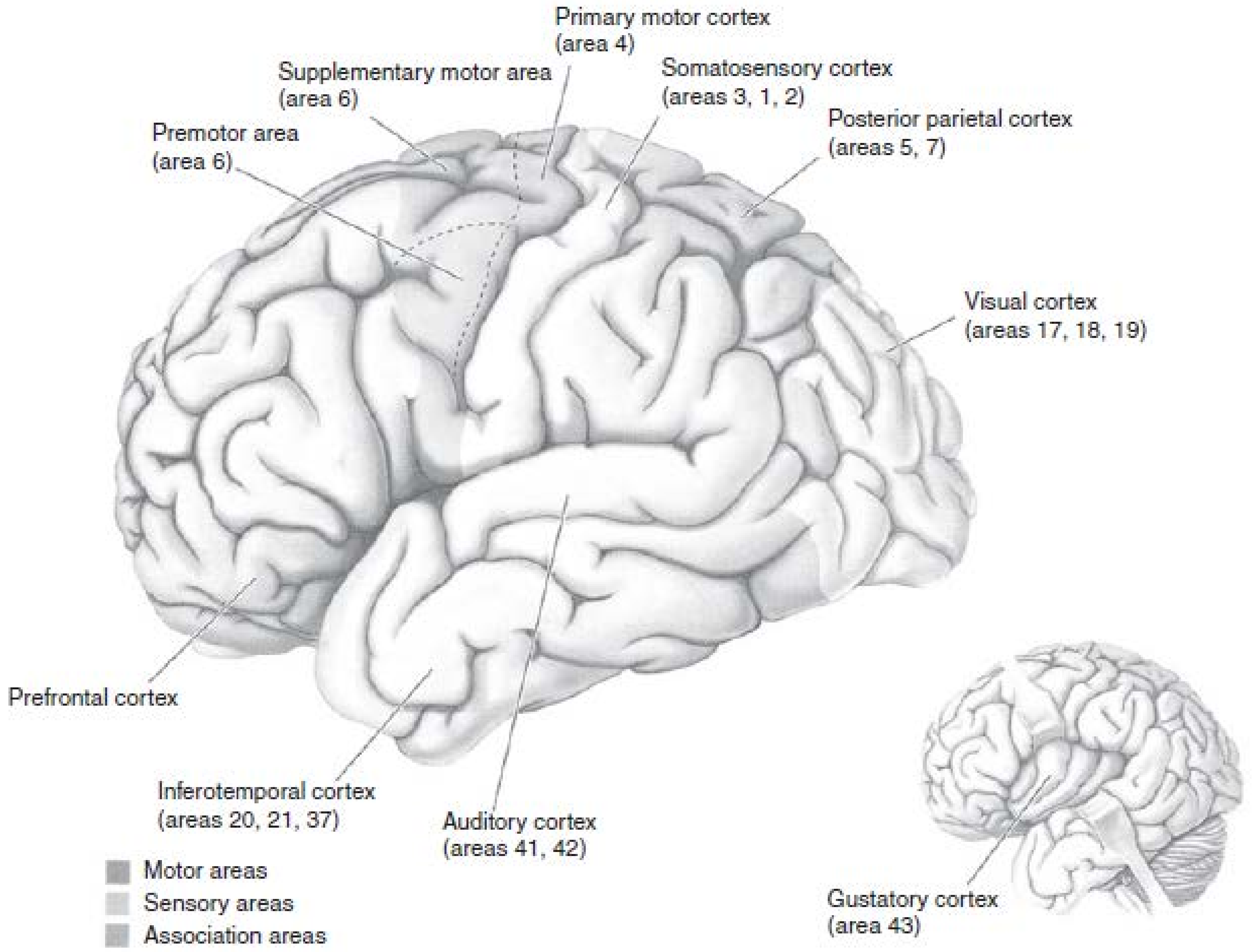
Forms of synaptic plasticity

- **LTP , LTD**
  - Numerous forms of synaptic plasticity have been experimentally observed, the most studied being long-term potentiation (**LTP**) and long-term depression (**LTD**). Both involve changes to a synapse that last for hours or even days.
- **STDP , STF/STD**
  - More recently, other types of plasticity have been characterized, including spike timing dependent plasticity (**STDP**), where the relative timing of input and output spikes determines the polarity of synaptic change, and short-term facilitation/depression (**STF/STD**), where the plasticity is rapid but not long-lasting.

2.7 Brain Organization, Anatomy, and Function

- **The design of a BCI** typically **involves** choices regarding **which brain areas** to **record from** and, in some cases, **which brain areas** to **stimulate**.
- This section provides **a brief overview** of brain **organization** and **anatomy**.


Thalamus
Cerebral Cortex
Pineal body
Hypothalamus
Tegmentum
Midbrain
Tectum
Pons
Medulla
Cerebellum


Primary motor cortex
(area 4)
Supplementary motor area
(area 6)
Somatosensory cortex
(areas 3, 1, 2)
Premotor area
(area 6)
Posterior parietal cortex
(areas 5, 7)
Visual cortex
(areas 17, 18, 19)
Prefrontal cortex
Inferotemporal cortex
(areas 20, 21, 37)
Auditory cortex
(areas 41, 42)
Gustatory cortex
(area 43)
Motor areas
Sensory areas
Association areas

2.8 Summary

- This chapter introduced the basic computing unit of the brain, the **neuron**, and the **brain anatomy**.

- We learned how neurons use electrical and chemical processes to communicate with one another, transmitting information "digitally" through **spikes** or **action potentials**.

- We also learned how such communication is mediated by junctions between neurons called **synapses**. Synapses can adapt at different time scales in response to inputs and outputs.

- **Long-term changes** in synaptic strength are thought to be the basis of memory and learning in the brain.

- We also learned the **anatomy** and **function area** of the brain.

2.8 Summary (Con.)

- As we shall see in subsequent chapters, **the fact** that information transmission in the brain is fundamentally electrical in nature **opens the door** to building a variety of BCIs that can record from and/or stimulate the brain.

- Additionally, the **plasticity** of the brain, as mediated by changes in synaptic strength, **plays a crucial role** in allowing novice BCI users to learn to **modulate** their brain activity in order to control novel devices.

Assignment

- Preview the next chapter.
- 3.1 Recording Signals from the Brain
  - 3.1.1 Invasive Techniques (Intracellular recording; Extracellular recording)
  - 3.1.2 Noninvasive Techniques (EEG; MEG; fMRI; fNIR; PET)
- 3.2 Stimulating the Brain
  - 3.2.1 Invasive Techniques (Microelectrodes; Direct Cortical Electrical Stimulation DCES; Optical Stimulation)
  - 3.2.2 Noninvasive Techniques (Transcranial Magnetic Stimulation TMS; Transcranial Ultrasound)
- 3.3 Simultaneous Recording and Stimulation
  - 3.3.1 Multielectrode Arrays
  - 3.3.2 Neurochip

- Any questions?